Mikhail Maslov, PhD

Quantitative Research | Systematic Strategies | Portfolio & Risk Modelling

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Theoretical physicist with experience in mathematical modelling, scientific computing, and machine learning. Seeking quant research roles focused on systematic strategies, portfolio and risk modelling. Willing to relocate Europe-wide.

EDUCATION

PhD in Theoretical Physics, Institute of Science and Technology Austria **(2018–2024)**
Thesis: *"Emergent physics of rotating quantum impurities in many-body environments"*
Research focus: group theory, mathematical physics, numerical modelling, and PINNs.

BSc in Mathematics and Physics, Moscow Institute of Physics and Technology **(2014–2018)**
Graduated with Honors (GPA 8.6/10), received Abramov and State Academic fellowships.

EXPERIENCE

Researcher, Institute of Science and Technology Austria **(2018–present)**
7 publications, 140 citations; reviewer for *Physical Review A* and *Physical Review Research*

- Built neural-network solvers for differential equations in quantum few-body systems using reinforcement learning and Monte Carlo sampling
- Developed a universal group-theoretical framework for analyzing angular momentum exchange in optical systems, applicable to many research fields in physics
- Developed a symmetry-aware mathematical model for optical sensing of material chirality
- Formulated analytical models for controlling quantum many-body systems; assessed their feasibility numerically using variational methods

Undergraduate Researcher, Moscow Institute of Physics and Technology **(2016–2018)**

- Developed a theoretical framework based on coupled partial differential equations to analyze plasmonic instabilities in nanostructured semiconductor systems
- Designed and maintained a MATLAB codebase for electrodynamics simulations of two-dimensional electron systems

Intern, Helmholtz Zentrum Dresden-Rossendorf **(summer 2017)**

- Designed and optimized code for molecular dynamics simulations on an HPC cluster

Intern, Deutsches Elektronen-Synchrotron, Hamburg **(summer 2016)**

- Developed MATLAB code for simulating crystallographic data using Monte Carlo method and comparing it to the experimental data from free-electron laser via Fourier analysis

Student Representative, Institute of Science and Technology Austria **(2019–2022)**

- Designed curriculum and taught atomic and condensed matter physics to 50+ PhD students
- Managed curriculum and student infrastructure development (500+ PhD students)
- Organized networking and scientific outreach events

Teaching assistant, Moscow Institute of Physics and Technology **(2014–2019)**

- Taught mathematics, physics and computer science to 100+ high-school students

SKILLS

Mathematics & modelling: linear algebra, tensor calculus, probability theory, differential equations, mathematical physics, group theory, differential geometry, quantum algorithms

Numerical methods & ML: Monte Carlo methods, variational methods, finite-difference methods, Fourier analysis, optimization, supervised and reinforcement learning

Programming: Python (NumPy, SciPy, PyTorch), MATLAB, Wolfram Mathematica, bash

Computing: vectorization, parallel computing, HPC, Linux, containerization, automation

Selected interests: game theory, decision theory, auction theory; market making and risk management; macroeconomics and public policy